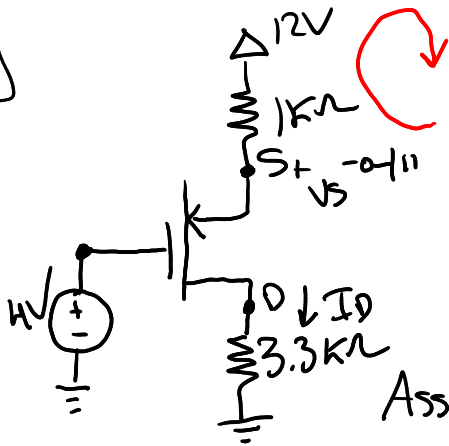


Final Exam Prep

Q1)



Given: $V_{TP} = -1.5V$ $K_p = 1.2 \frac{mA}{V^2}$

Find: V_S, V_{SG}, I_D, V_D , and verify whether in triode or Saturation

$$-V_S - 1k\Omega I_D + 12 = 0 \Rightarrow V_S = 12 - 1000 I_D \quad (1)$$

$$V_{SG} = V_S - V_G \Rightarrow V_{SG} = 12 - 1000 I_D - 4$$

$$\Rightarrow V_{SG} = 8 - 1000 I_D$$

Assume SAT:

$$I_D = \frac{1}{2} K_p (V_{SG} - |V_{TP}|)^2$$

$$I_D = 0.6 \times 10^{-3} (8 - 1000 I_D - 1.5)^2 \Rightarrow I_D = 0.6 \times 10^{-3} (6.5 - 1000 I_D)^2$$

Solve for I_D

$$I_D = 0.6 \times 10^{-3} (42.25 - 13,000 I_D + 1 \times 10^6 I_D^2)$$

$$\Rightarrow \frac{I_D}{-I_D} = 25.35 \times 10^{-3} - 7.8 I_D + 600 I_D^2$$

$$\Rightarrow 600 I_D^2 - 8.8 I_D + 25.35 \times 10^{-3} = 0$$

$$I_D = 10.7 mA \text{ or } I_D = 3.94 mA$$

$$V_{SG} \Rightarrow 8 - 1000 (10.7 \times 10^{-3}) \text{ or } V_{SG} = 8 - 1000 (3.94 \times 10^{-3})$$

$$V_{SG} = -2.7V$$

$$V_{SG} > |V_{TP}|$$

$$-2.7 > 1.5V$$

NO

$$4.06 > 1.5$$

Yes

$$V_S = 12 - 1000 (3.94 \times 10^{-3}) = 8.06V$$

$$V_D = I_D R_D = 13V$$

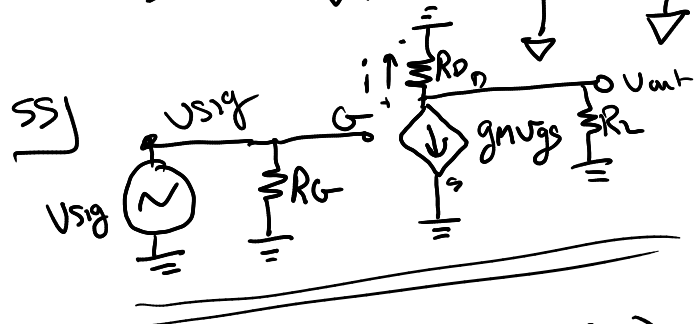
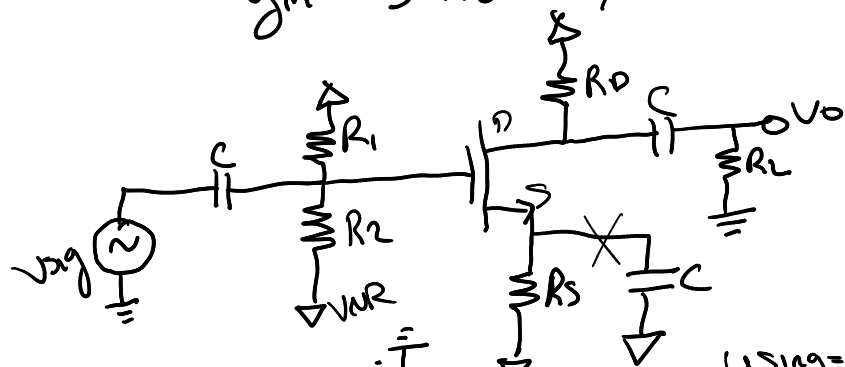
$$V_{SD} = V_S - V_D = -4.94$$

$$|V_{SD}| > V_{SG} - |V_{TP}| \Rightarrow 4.94 > 4.06 - 1.5$$

Proves Saturation

Q2 An NMOS is biased in Saturation where
 $g_m = 5 \text{ mS}$, $R_D = 6.8 \text{ k}\Omega$, $R_L = 10 \text{ k}\Omega$

- 1) Draw the small signal model
- 2) Determine the effective output resistance seen at the drain
- 3) Calculate the voltage gain



Using Hybrid- π

$$R_{eq} = R_D || R_L = 6.8 \text{ k}\Omega || 10 \text{ k}\Omega = 4.05 \text{ k}\Omega$$

$$A_v = \frac{V_{out}}{V_{sig}} = \frac{V_{out}}{V_{gs}}$$

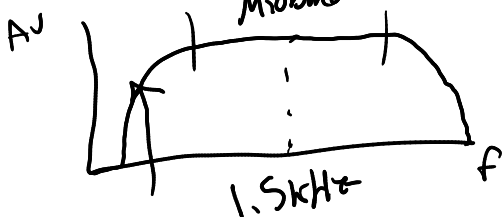


$$V_{out} = -g_m V_{gs} (R_{eq})$$

$$V_{gs} = V_g - V_s \Rightarrow V_g = V_{sig} \quad V_s = V_{signal}$$

$$A_v = \frac{-g_m V_{sig} (R_{eq})}{V_{sig}}$$

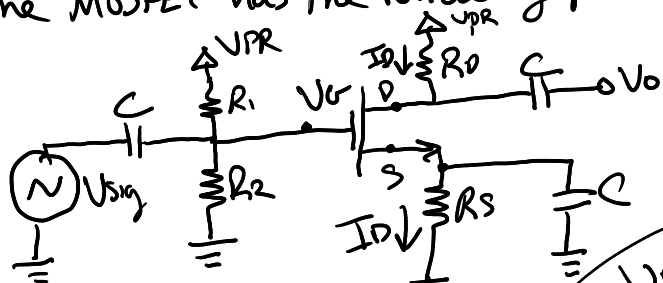
$$\Rightarrow A_v = -g_m R_{eq} \text{ or } A_v = -g_m (R_D || R_L) = -20.25 \frac{V}{V}$$



Q3 Design a common-source NMOS amplifier that satisfies
 $V_{DD} = 15 \text{ V}$, $I_D = 2 \text{ mA}$, $V_D = 7.5 \text{ V}$, $V_{DS} = 5 \text{ V}$, and $|A_v| = 18$

Also choose R_1 and R_2 such that $R_1 + R_2 = 20 \text{ k}\Omega$

The MOSFET has the following parameters: $V_{TN} = 2 \text{ V}$, $K_n = 1.5 \frac{\text{mA}}{\text{V}^2}$



Determine R_D , R_S , g_m , V_{GS} , R_1 and R_2

For small signal gain, $A_v = -g_m R_D$

$$|g_m| = \frac{-18}{R_D} = \frac{-18}{3.75 \text{ k}\Omega} \approx 4.8 \text{ mS}$$

$$V_{DS} = 5 \text{ V and } V_{DS} = V_D - V_S \Rightarrow V_S = 7.5 - 5 = 2.5 \text{ V}$$

$$R_D = \frac{15 - 7.5}{2 \text{ mA}} = 3.75 \text{ k}\Omega$$

$$R_S = \frac{V_S}{I_D} = 1.25 \text{ k}\Omega$$

$$g_m = \frac{2I_D}{V_{GS} - V_{TN}} \Rightarrow V_{GS} - V_{TN} = \frac{2I_D}{g_m} \Rightarrow V_{GS} = \frac{2I_D}{g_m} + V_{TN} = 2.93 \text{ V}$$

$$V_{GS} = V_G - V_S \Rightarrow V_G = V_S + V_{GS} = 5.33 \text{ V}$$



$$V_G = \frac{R_2}{R_1 + R_2} V_{DD} \Rightarrow \frac{R_2}{20 \text{ k}\Omega} (15) = 5.33 \text{ V}$$

$$R_2 = 7.11 \text{ k}\Omega \text{ and } R_1 = 12.89 \text{ k}\Omega$$